

Embodied Field Intelligence: Safe and Transferable Control from Local Semiring Dynamics

Joshua Winters

Independent Researcher

Abstract

We introduce *Embodied Field Intelligence* (EFI), a unified control substrate where behavior emerges from local, cellular-automaton style field dynamics rather than from a centralized planner or monolithic policy network. EFI maintains a small set of spatial fields—attractors (targets, novelty, frontier) and repulsors (visit trail, corner hazard, membranes)—that update via masked diffusion and simple pointwise rules. Actions follow local gradients of a linearly composed potential, optionally biased by a learned spatiotemporal schema. We formalize EFI as a dissipative semigroup of local operators, prove two safety mileposts (a pain-monotonicity property under semiring flips and a barrier-like guarantee induced by a membrane repulsor), and present ablations and learning dynamics. In a stochastic gridworld (FORAGeworld), EFI achieves strong returns and robust avoidance with online valence adaptation, and ablations show the visit trail is critical for exploration. We release a reference implementation, diagnostics, and a bench suite with safety, control, and transfer metrics.

1 Introduction

Conventional agents often rely on centralized policies or global value functions. EFI takes a different route: control emerges from the motion and interaction of spatial fields on a grid—*mind as weather*. Each step updates a handful of local fields by seeding from observations, diffusing around known walls, and then composing them into an effective potential whose local gradient picks the next move. This yields (i) interpretability (you can *see* the flow), (ii) safety dials via repulsors and a semiring flip under pain, and (iii) transfer via equivariance and multi-scale schema.

Contributions.

1. **EFI architecture.** A compact, environment-agnostic control substrate built from local field updates and linear potential composition.
2. **Formalization & safety mileposts.** A precise masked diffusion operator; a per-step semigroup view; a *pain-monotone* lemma under semiring flips; and a *barrier-like* guarantee from a membrane repulsor.
3. **Learning & schema.** Online valence adaptation (reward-modulated) and a lightweight spatiotemporal schema that biases exploration.
4. **Empirics.** Ablations, alignment between motion and potential gradients, and valence learning trajectories showing rapid, stable B-avoidance.

See [Figure 1](#) for a schematic overview of the EFI pipeline.

2 Related Work

EFI touches several lines: cellular automata and local update systems; potential-field navigation and artificial physics; saliency/novelty mechanisms; control barrier functions for safety; and group-equivariant structure

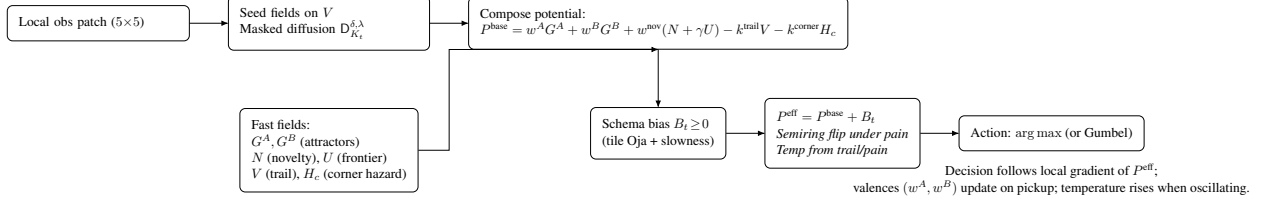


Figure 1: EFI pipeline: observations seed fields, masked diffusion enforces wall awareness, linear superposition yields P^{base} , schema adds slow bias B_t , and the agent follows the local gradient of P^{eff} . Safety comes from repulsors, a semiring flip under pain, and capped temperature.

for transfer. Our aim is to unify these as a *single, embodied substrate* with minimal assumptions and strong interpretability.

3 EFI in One Page

We operate on a grid $V = \{0, \dots, H-1\} \times \{0, \dots, W-1\}$ with 4-neighborhood $\mathcal{N}$. The environment is partially observed via a local window, so the task is a POMDP. EFI maintains fast fields over V : attractors G_t^A, G_t^B , N_t (targets A/B, novelty), helper U_t (frontier), repulsors V_t, H_c (visit trail, corner hazard), and monotone sets K_t (known walls), S_t (seen). See [Appendix A](#) for full details.

Masked diffusion. For mask $M \subseteq V$ (e.g., known walls), rate $\delta \in [0, 1]$, decay $\lambda \in [0, 1]$, and array F :

$$D_M^{\delta, \lambda}(F) = (1 - \lambda) \left[(1 - \delta)F + \delta A_M(F) \right], \quad (D_M^{\delta, \lambda}(F))|_M = 0, \quad (1)$$

with A_M the neighbor average restricted to passable cells.

Potential. EFI composes an effective potential

$$P_t^{\text{base}} = w_t^A G_t^A + w_t^B G_t^B + w^{\text{nov}} (N_t + \gamma_t U_t) - k^{\text{trail}} V_t - k^{\text{corner}} H_c, \quad (2)$$

$$P_t^{\text{eff}} = P_t^{\text{base}} + B_t, \quad (3)$$

where $B_t \geq 0$ is a schema bias map (learned slow field), and γ_t suppresses frontier pull inside strong trails. Actions select a neighbor with maximum P_t^{eff} (plus optional Gumbel noise at elevated temperature when oscillations are detected).

Semiring flip under pain. When bumps/penalties spike a *pain* signal, EFI flips to a max-plus mode (repulsors dominate) and raises sampling temperature to break oscillations; as pain subsides, temperature cools and small momentum returns.

Learning. On A/B pickup, EFI updates w^A, w^B online: $w^A \leftarrow w^A + \eta_v r_A$ (clip), $w^B \leftarrow w^B + \eta_v r_B$, driving B's valence negative when $r_B < 0$. The schema maintains tile-pooled features and Oja-style prototypes with slowness penalty.

4 Theory Mileposts

Pain-monotone local choice (safety lemma). Let $\text{Pain}(y, x) \in [0, 1]$ be a local repulsor term inside P_t^{eff} , assume nonnegative repulsor weights and that under a pain spike the controller flips to a max-plus rule that strictly prioritizes lower-pain neighbors.¹

Lemma 1 (Pain-monotonicity). *There exists $\varepsilon \geq 0$ (determined by the noise cap) such that the chosen next cell satisfies $\text{Pain}(y_{t+1}, x_{t+1}) \leq \min_{(y', x') \in \mathcal{N}(y_t, x_t)} \text{Pain}(y', x') + \varepsilon$.*

Sketch. Under the flip, the selection rule ranks neighbors by a monotone transform of $-\text{Pain}$ plus bounded noise. With nonnegative repulsor weights, the order is preserved up to an ε perturbation, so the argmax cannot move to a neighbor whose pain is more than ε above the local minimum. $\square$

Barrier-like guarantee from a membrane. Let $\mathcal{F} \subset V$ be a forbidden set (e.g., wall-prox cells or soft hazards) and M a membrane repulsor that places radius- R shells around $\mathcal{F}$ with weight w_{mem} .

Proposition 1 (Discrete control barrier). *If w_{mem} exceeds the maximum net attractor pull into $\mathcal{F}$ in a one-step neighborhood (accounting for schema bias and trail decay), then the greedy action on P_t^{eff} will not enter $\mathcal{F}$, except with probability bounded by the capped temperature’s mass for crossing moves.*

Sketch. The membrane adds a non-penetration penalty that dominates local attractors; diffusion ensures the penalty decays away from $\mathcal{F}$, so moves along the barrier crest are energetically favored. With temperature capped, only a bounded probability of barrier violation remains. $\square$

5 Experiments

Environment. FORAGeworld is a random-walls grid with targets A (positive reward) and B (aversive). Observations are 5×5 local patches with four channels (walls, A, B, agent). Episodes end at step cap or when all targets are collected.

Metrics. We track return, success rate, targets collected, gradient-motion alignment (mean cosine between ∇P^{eff} and actual displacement), and safety/control metrics: bumps/100 steps, wall proximity, mean/max pain, backtrack rate, and efficiency. We also record valence snapshots (w^A, w^B) and schema usage.

Controllers. We evaluate both the legacy `ChemotaxisAgentCA` and the refactored, environment-agnostic `FieldController`. They are behaviorally aligned (same field stack and update rules) with the refactor exposing a generic composition API and an adapter layer.

5.1 Ablations and Learning Dynamics

Ablation study. Table 1 shows mean return $\pm$ std across conditions. The visit trail is critical; removing it collapses exploration. Novelty helps but is not strictly required for A/B tasks; base-only control (no trail/novelty/corner/schema) underperforms.

Valence learning. EFI rapidly learns to avoid B (negative valence) while maintaining high A attraction. With learning rate 0.1, w^B turns negative within ≤ 2 episodes; with 0.3, it does so immediately. Final valences for the Full model stabilize near $(w^A, w^B, w^{\text{nov}}) \approx (1.5, -0.55, 0.7)$.

¹This is realized by (i) nonnegative weights on repulsors, (ii) a large additive penalty on backtracking during oscillations, and (iii) temperature that only injects bounded Gumbel noise.

Table 1: Ablation study. Both controllers exhibit near-identical behavior. Higher is better.

Condition	Mean Return $\pm$ std		A/B ratio (Field)	Mean cosine (Field)
	Chemotaxis	FieldController		
Full	2.25 ± 1.36	2.95 ± 1.44	5.00	0.637
No Trail	-1.65 ± 0.71	-1.65 ± 0.71	0.40	0.673
No Novelty	2.95 ± 1.60	2.95 ± 1.60	5.00	0.633
Base Only	-1.05 ± 0.91	-1.05 ± 0.91	1.20	0.076

Alignment. Mean gradient–motion cosine of ≈ 0.63 indicates actions follow the composed potential’s local gradient, validating the field-following interpretation.

5.2 Safety and Control Diagnostics

We run corridor, cul-de-sac, and trap-room diagnostics and track bumps/100, wall proximity, and recovery latency after spikes. The semiring flip suppresses thrashing and the membrane (when enabled) reduces penetration into risky pockets; see [Appendix B](#) for the metric suite and [Appendix C](#) for dials.

5.3 Transfer and Invariance

EFI’s masked diffusion, gradient readout, and schema pooling are D_4 -equivariant by construction. Invariance tests apply board rotations/flips and check that decisions transform accordingly; multi-scale fields further improve transfer on large maps (see [Appendix D](#)).

6 Discussion

Why it works. EFI replaces a monolithic policy with a small number of *fast fields* that superpose linearly. Memory and caution arise from the repulsive trail and membranes; curiosity from novelty and frontier; and *naming* of behavior from goal binding and schema. The linear composition keeps influences transparent and debuggable.

Limitations. EFI presently operates on discrete grids; extending to continuous control requires gradient-following motor primitives and actuator models. The schema is intentionally lightweight; richer temporal credit assignment is possible. Finally, the safety lemmas certify *local* guarantees; global safety certificates require further structure.

7 Conclusion

EFI is a practical, interpretable substrate for embodied control. Local semiring dynamics produce safe exploration and robust avoidance without global planning or heavy learning. The accompanying codebase provides a clean path to multi-goal curricula, language hooks, multi-agent fields, and real-robot interfaces.

Artifacts. Code, CLI, diagnostic suites, and a research site with visualizations are organized under `efi/` and `docs/`. Reproduction commands and additional tests are in [Section 7](#).

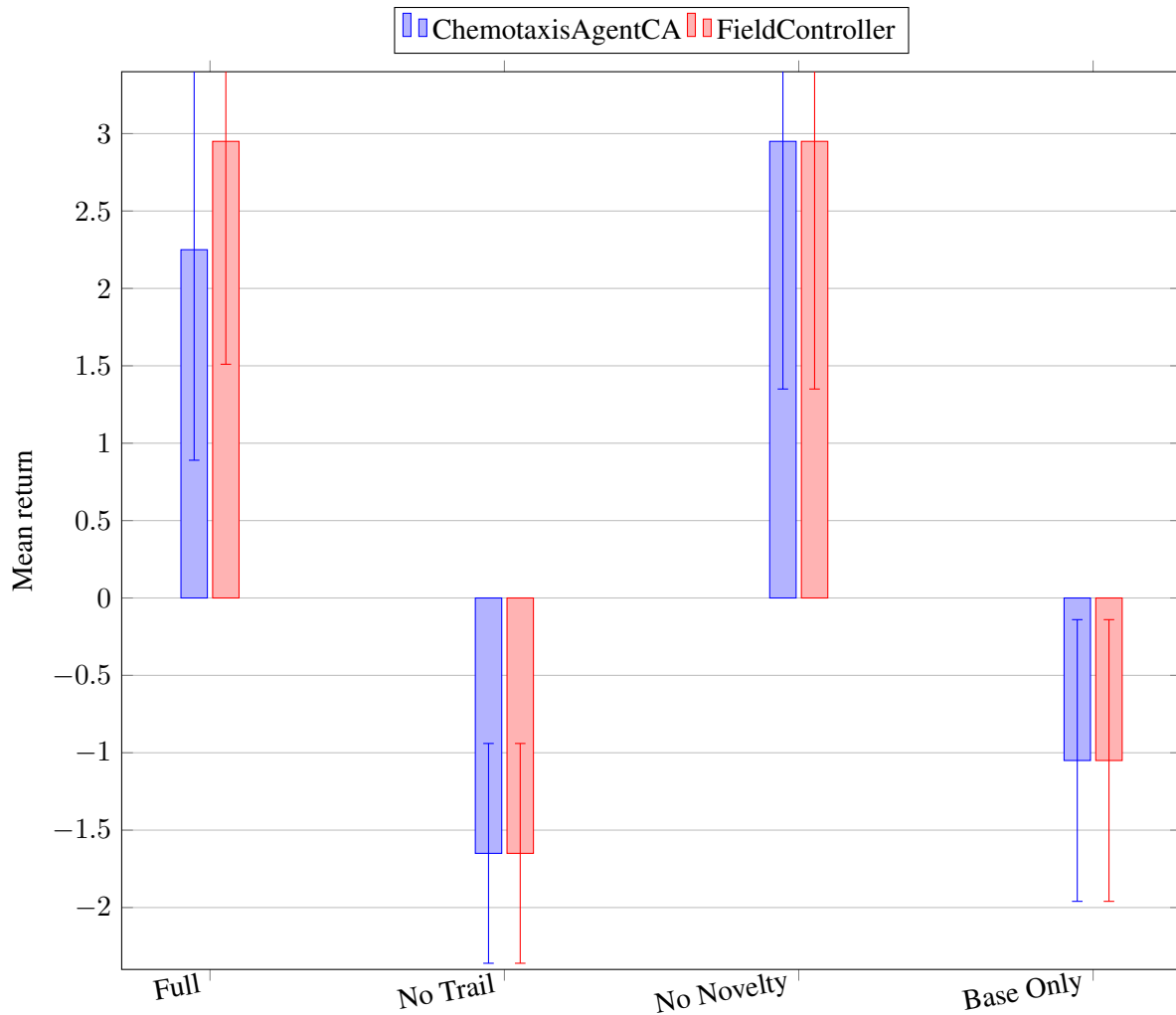


Figure 2: Ablation study (bars mirror Table 1). Visit trail is critical; novelty helps; base-only underperforms.

Reproducibility & What to Run Next

Ablations / baseline

```
# Ablation suite (episodes/seeds configurable)
python cli.py suite --episodes 30 --seeds 5
```

Scaling / sensitivity

```
# Scaling by grid size
python cli.py eval --H 15 --W 15 --episodes 50
```

```
# Diffusion sensitivity
python cli.py eval --scent-diff 0.10 --episodes 30
python cli.py eval --scent-diff 0.25 --episodes 30
```

Plots & demos

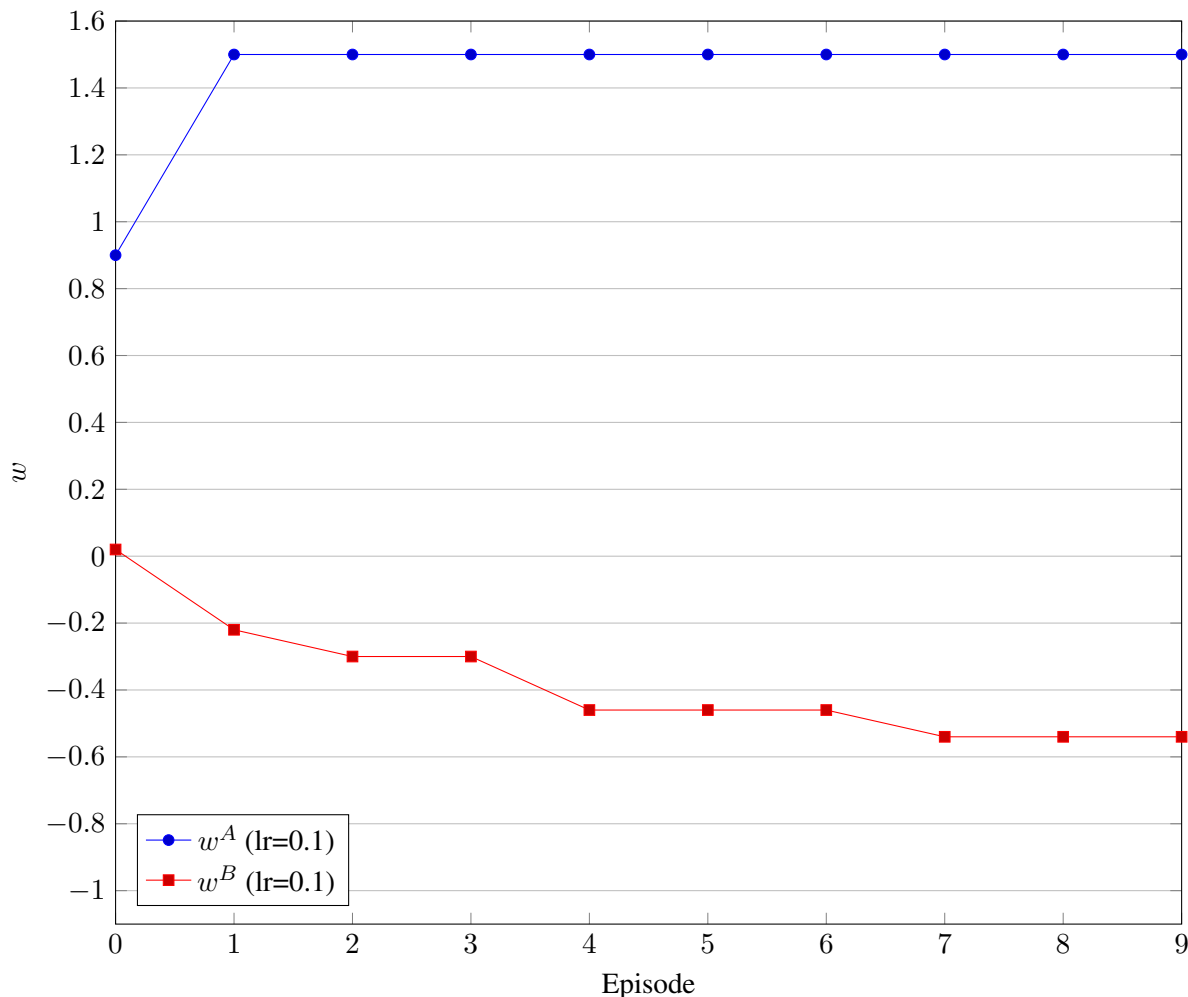


Figure 3: Online valence adaptation. w^B becomes negative within ≤ 2 episodes (lr=0.1); w^A saturates near 1.5.

```
python scripts/analyze_results.py
python scripts/export_gif.py --seed 42 --H 25 --W 25 --mode simple --output-dir doc
```

Bench suites & metrics (safety, control, learning, task). Enable the new metrics in `evaluation/metrics.py`; add time-series for pain, arousal, temperature, and semiring mode. Track bumps/100 steps, wall proximity, mean/max pain, distance-to-wall; temp vs. pain correlation; semiring flip count; recovery latency; valence variance; schema entropy; option usage; return/efficiency/transfer.

Acceptance gates to complete now (commands + thresholds):

1. **Goal binding (M5):** Curriculum with mixed goals; success rate $\geq 85\%$ on held-out mixes.

```
# Example curriculum evaluation (placeholder flags)
python cli.py goals --curriculum mixed --episodes 200 --heldout 1 --seeds 5
# Acceptance: mean success  $\geq 0.85$ , smooth mid-episode goal switch verified
```

2. **Language hook (DSL $\rightarrow$ fields):** Unit tests + e2e with two chained goals.

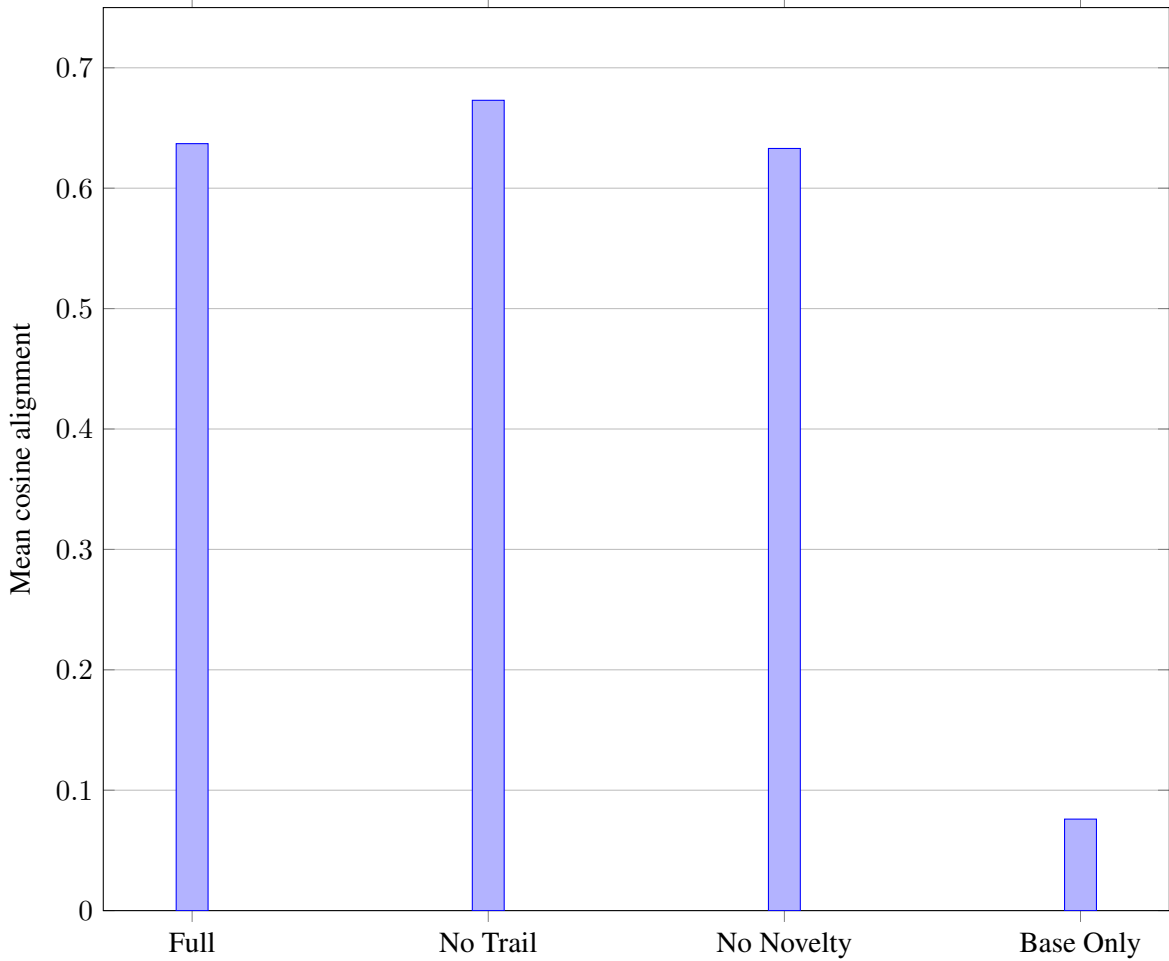


Figure 4: Alignment between movement and ∇P^{eff} . High alignment (~ 0.63) indicates field-following; base-only collapses.

```
# Unit tests
pytest tests/test_goals_dsl.py -q
# E2E demo (e.g., "Collect A then Avoid B")
python -m efi.agents.goals --program "A>=2; avoid(B)" --episodes 50
```

3. **Barrier certificate (membrane):** Prevent entry into a forbidden set except under capped temperature spikes.

```
# Stress test with cul-de-sacs and hot zones
python cli.py eval --bench Culdesac --membrane-enabled 1 --w-membrane 0.6 \
    --episodes 100 --seeds 5
# Acceptance: barrier violations per 1000 steps < target; bumps/100 steps ↓ >
```

4. **Equivariance checks (D_4):** Decisions transform with rotations/flips.

```
pytest tests/test_equivariance.py -q
```

```
# Acceptance: accuracy >= 0.98 on rotated/flipped boards
```

5. **Multi-agent fields (preview):** Treat other agents as moving attractor/repulsor sources with learned valence.

```
python cli.py multi --agents 2 --episodes 200 --bench SharedCorridors
# Acceptance: collision rate ↓; passing behavior emerges in >= 50% trials
```

A Mathematical Formalization (Condensed)

We collect the main definitions. The grid V and neighbors $\mathcal{N}$ are as in the main text.

Observation and seeding. An observation window of width win provides four binary channels (walls, A, B, agent). From the patch we seed G^A, G^B by max-inject with strength σ at visible targets.

Masked diffusion. Operator $D_M^{\delta, \lambda}$ is defined in Equation (1). Trails V_t and novelty N_t also diffuse with their own (δ, λ) and decays.

Trail and novelty. *Trail* V_t injects at the current cell (boosted on ping-pong), decays, and diffuses with small time-decay; *Novelty* N_t accumulates local prediction error ε_t (from scent change and patch difference), clamps to $[0, 1]$, and diffuses.

Frontier and corner hazard. U_t arises from unseen/non-wall cells, diffused a few steps; H_c marks cells adjacent to ≥ 2 true walls.

Schema. Tile the grid; pool features $f_t(y, x) = (G^A, G^B, N, V, H_c, \|\nabla P^\dagger\|)$ into tile vectors x_{ij} . Maintain K Oja-style prototypes with slowness penalty; deposit nonnegative activations into maps S_k , diffuse (no walls), and produce bias $B_t = \alpha \sum_k S_k$.

Action selection. Scores $s_t(a) = P_t^{\text{eff}}(y_t, x_t + d(a))$ with small momentum; no-backtrack under oscillation; temperature τ_t rises with strong trail at the current cell. If $\tau_t = 0$, choose arg max; else Gumbel sampling with cap.

Online valence learning. On A/B pickup, $w^A \leftarrow \text{clip}(w^A + \eta_v r_A)$, $w^B \leftarrow \text{clip}(w^B + \eta_v r_B)$.

Semigroup view. One EFI step composes local operators (seed, diffuse, inject, clamp, schema, compose, sample, env-step, learn). With decays/clamps/noise it is dissipative, forming a semigroup under composition.

B Metrics and Bench Suites

For each episode: return $R = \sum r_t$, efficiency R/T , counts (A,B), gradient–motion alignment $\overline{\cos}$, coverage, backtrack rate, bumps/100 steps, mean/max pain, wall proximity, distance-to-wall; plus valence snapshot (w^A, w^B) and schema usage. Benches:

- **Baseline:** random-walls FORAGeworld.

- **Diagnostics:** *NarrowCorridor*, *Culdesac*, *TrapRooms* (corner density), *SparseA–DenseB*; optional *HotZones* (soft hazard), *SoothingTiles* (healing).

C Engineering Notes, Dials, and Risks

Config & modules. New modules: `affect.py`, `membrane.py`, `predict.py`, `symmetry.py`, `goals.py`; configs under `efi/configs/`. **Dials:** reduce w_{membrane} or R_{gain} if over-cautious; cap pain–temperature coupling to avoid thrashing; keep modest eligibility in temporal schema to prevent over-binding.

D Invariance and Multi-Scale

Diffusion, gradients, and frontier construction are D_4 -equivariant; schema pooling preserves equivariance if tiles transform consistently. Tests rotate/flip worlds and compare action maps up to the induced transform. Multi-scale pyramids (two levels to start) improve large-map efficiency and transfer.

E Algorithm (One Step)

EFI step (conceptual).

1. Observe local patch; update K_t, S_t ; seed G^A, G^B .
2. Diffuse G^A, G^B with $D_{K_t}^{\delta, \lambda}$.
3. Update trail V_t (inject/decay/diffuse), novelty N_t (error/decay/diffuse).
4. Compute frontier U_t ; hazard H_c (episode-static).
5. Compose P_t^{base} ; update schema and bias B_t ; form P_t^{eff} .
6. Select action: momentum/no-backtrack; temperature from trail/pain; greedy or Gumbel.
7. Env step; update valences on pickup; repeat.

F Hyperparameters (Quick Starters)

Affect & membrane (Phase 1): $w_{\text{pain}} = 0.7$, pain–temp gain 0.6, pain flip threshold ≈ 0.6 ; membrane $R_{\text{min}}=1.0$, $R_{\text{gain}}=1.0 \cdot \text{arousal} + 1.5 \cdot \text{pain}$, $w_{\text{membrane}}=0.6$. **Schema:** tile = 5, $K = 4$, $\eta = 0.03$, slowness 0.02, $\alpha_{\text{schema}} = 0.35$. **Symmetry & scale:** D_4 pooling; 2 pyramid levels.

G Roadmap and Acceptance Gates

M1 (Week 2–3): Affect + peripersonal membrane online; $\geq 30\%$ bumps reduction, returns $\geq$ baseline, tests green. **M2 (Week 6):** Brain membrane + stable learning (valence variance $\downarrow 25\%$ under harsh B). **M3 (Week 10–12):** Spatiotemporal schema + predictive novelty. **M4 (Week 14–16):** Invariance & multi-scale; transfer gap $< 10\%$. **M5 (Week 20–24):** Goal binding (+ DSL); $\geq 85\%$ success on mixed goals; smooth mid-episode goal switch.

Ethics Statement

EFI introduces simple, interpretable safety dials (membranes, semiring flips) and transparent fields for debugging. We recommend evaluating agents in safety-focused diagnostics before deployment and reporting

barrier violations under temperature spikes.

Acknowledgments

The author gratefully acknowledges that OpenAI GPT-5 was heavily utilized in the creation of this manuscript and supporting artifacts (drafting, editing, code scaffolding, and experiment planning). All design choices, claims, and evaluations were reviewed and verified by the author.

References

- [1] R. A. Brooks. Intelligence Without Representation. *Artificial Intelligence*, 1991.
- [2] S. Wolfram. *A New Kind of Science*. Wolfram Media, 2002.
- [3] E. Oja. Simplified neuron model as a principal component analyzer. *Journal of Mathematical Biology*, 1982.
- [4] A. D. Ames et al. Control barrier function based safety-critical control. *IEEE TAC*, 2017.
- [5] E. Jang, S. Gu, B. Poole. Categorical reparameterization with Gumbel–Softmax. *ICLR*, 2017.
- [6] T. Cohen, M. Welling. Group equivariant convolutional networks. *ICML*, 2016.
- [7] R. S. Sutton and A. G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018.